Primary Servo Capstan (S.O.O. 6221) – Maintenance Practices

1. Removal/Installation

A. Primary Servo Capstan (Rudder, Elevator and Aileron)

For Rudder refer to Figure 1 for details.

For Elevator refer to Figure 2 for details.

For Aileron refer to Figure 3 for details.

1) Removal of Primary Servo Capstan

a) Ensure the aircraft electrical power is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Remove locking clips and unscrew turnbuckle to release servo bridle cable tension.

c) Disconnect cable clamps.

d) Disconnect electrical connectors from servos.

e) Supporting the servo and servo capstan, remove bolts securing servo capstan bracket to aircraft structure, disengage capstan bracket from mounting.

Note Inspect bracket mounting pads and mounting structure foot print for corrosion.

2) Installation of Primary Servo Capstan

WARNING INSTALLATION OF THIS COMPONENT CONTAINS PROTECTION FEATURES THAT ARE CRITICAL TO PREVENTING DEGRADATION FROM THE EFFECTS OF HIGH INTENSITY RADIATED FIELDS (HIRF) AND INDIRECT EFFECTS OF LIGHTNING (IEL). ENSURE INTEGRITY OF THE HIRF/IEL PROTECTION FEATURES ARE MAINTAINED.

a) Ensure the aircraft electrical power is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Servo to airframe electrical bonding is maintained through direct contact between the servo and airframe mating surfaces. Prior to installation, ensure mating surfaces are clean and free of contaminants. Refer to procedures in 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data, as required.

c) Ensure capstan has been correctly rigged.

d) Mount capstan to supporting aircraft structure.

e) Torque mounting bolts to 55±5 in-lbs above running torque.

f) Connect servo cable to turnbuckle and to clamp. (Refer to Figure 1, Figure 2 and Figure 3).

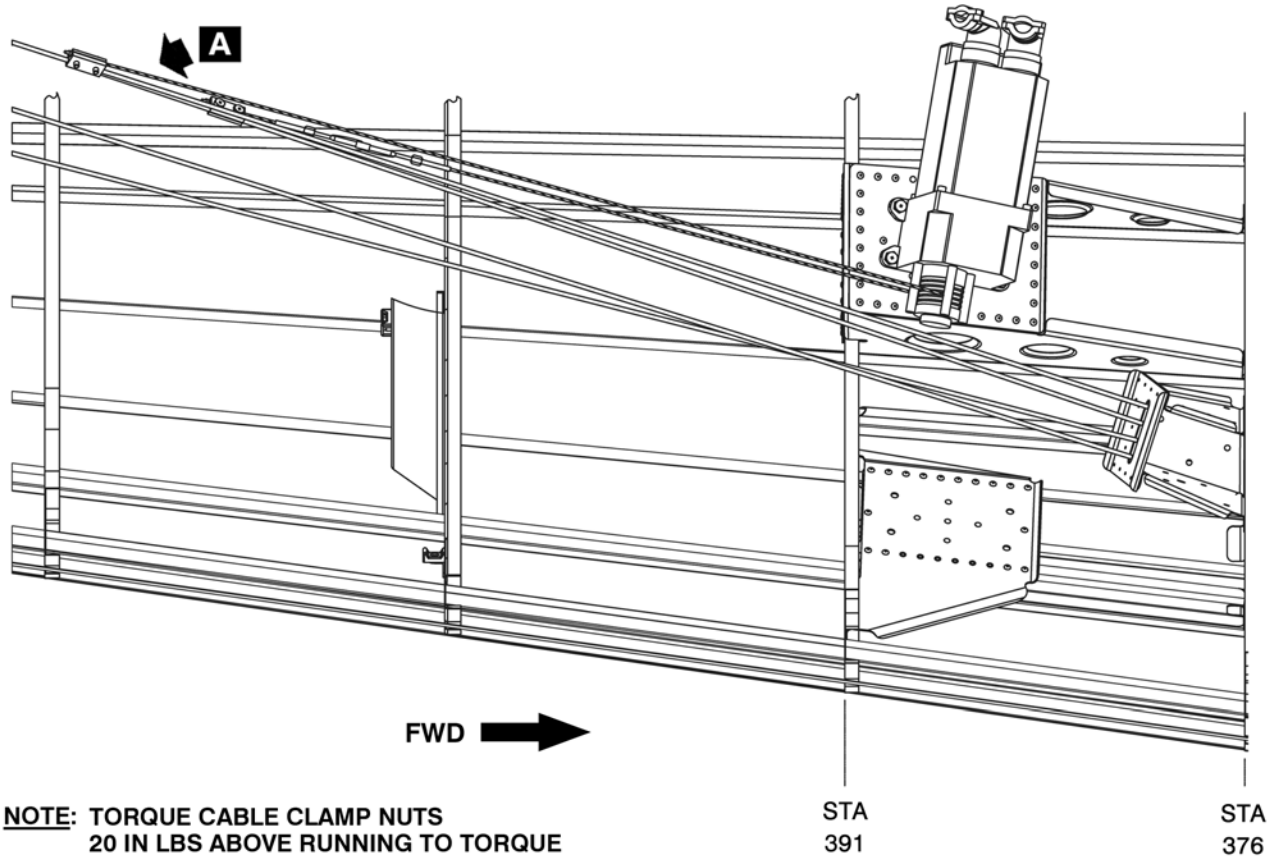
g) Torque clamp nuts to 20 in-lbs above running torque.

h) Rig bridle cables in accordance with Para 2.

i) Connect electrical connectors to servos.

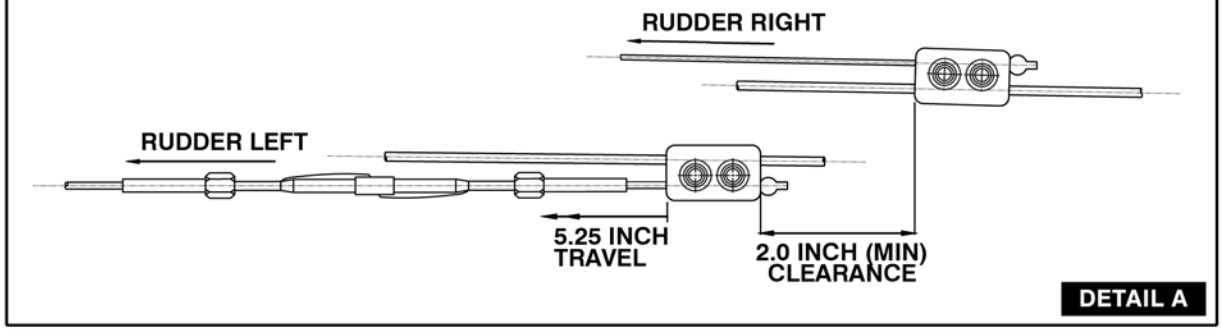
j) Do bonding check. (Refer to 22-10-11, Primary Servos (S.O.O. 6221) – Maintenance Practices, Para 3.A.).

VIEW LOOKING INBOARD AND DOWN ON RIGHT SIDE
SHOWING INSTALLATION OF RUDDER SERVO
(ELEVATOR SERVO OMITTED FOR CLARITY)



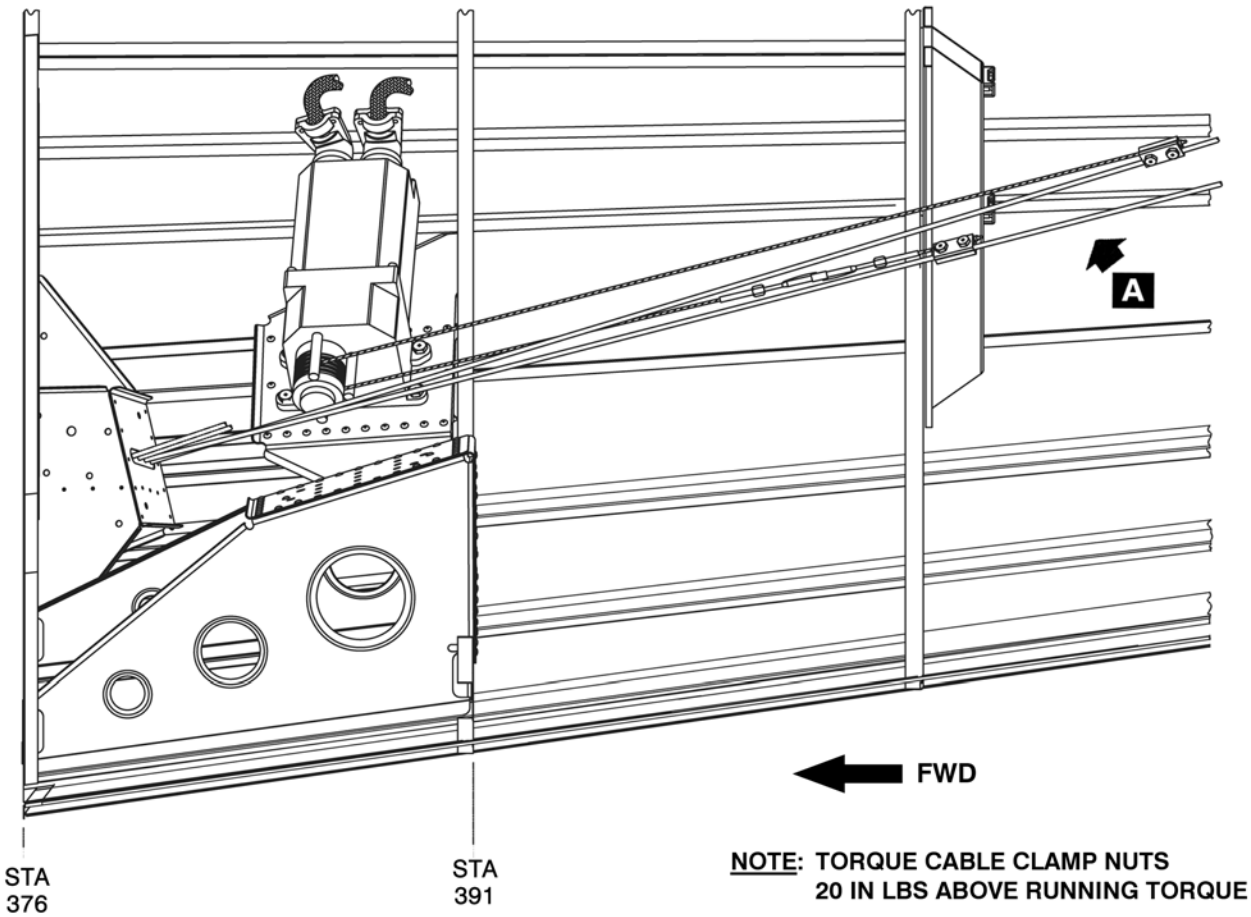
NOTE: TORQUE CABLE CLAMP NUTS
20 IN LBS ABOVE RUNNING TO TORQUE

SHOWING SECURING OF RUDDER SERVO BRIDLE CABLES TO RUDDER CONTROL CABLES
RUDDER IN FULL RIGHT POSITION (RIGHT RUDDER PEDAL FULL FORWARD)

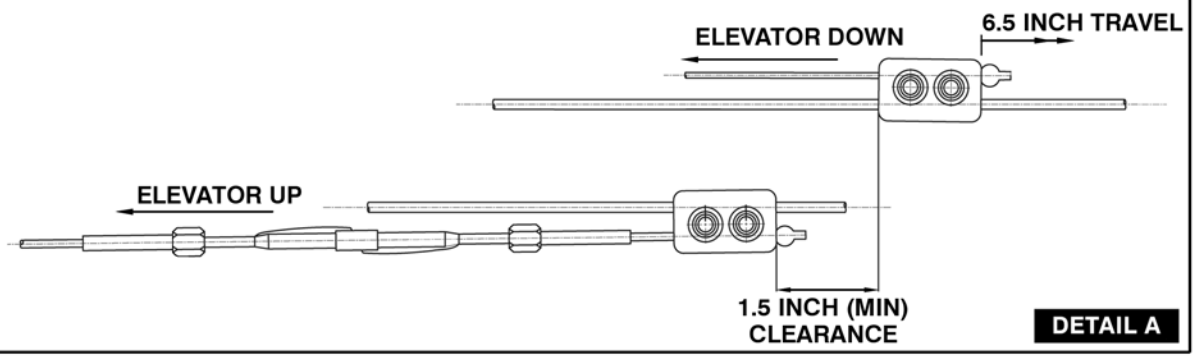


Rudder Servo Capstan Installation
Figure 1

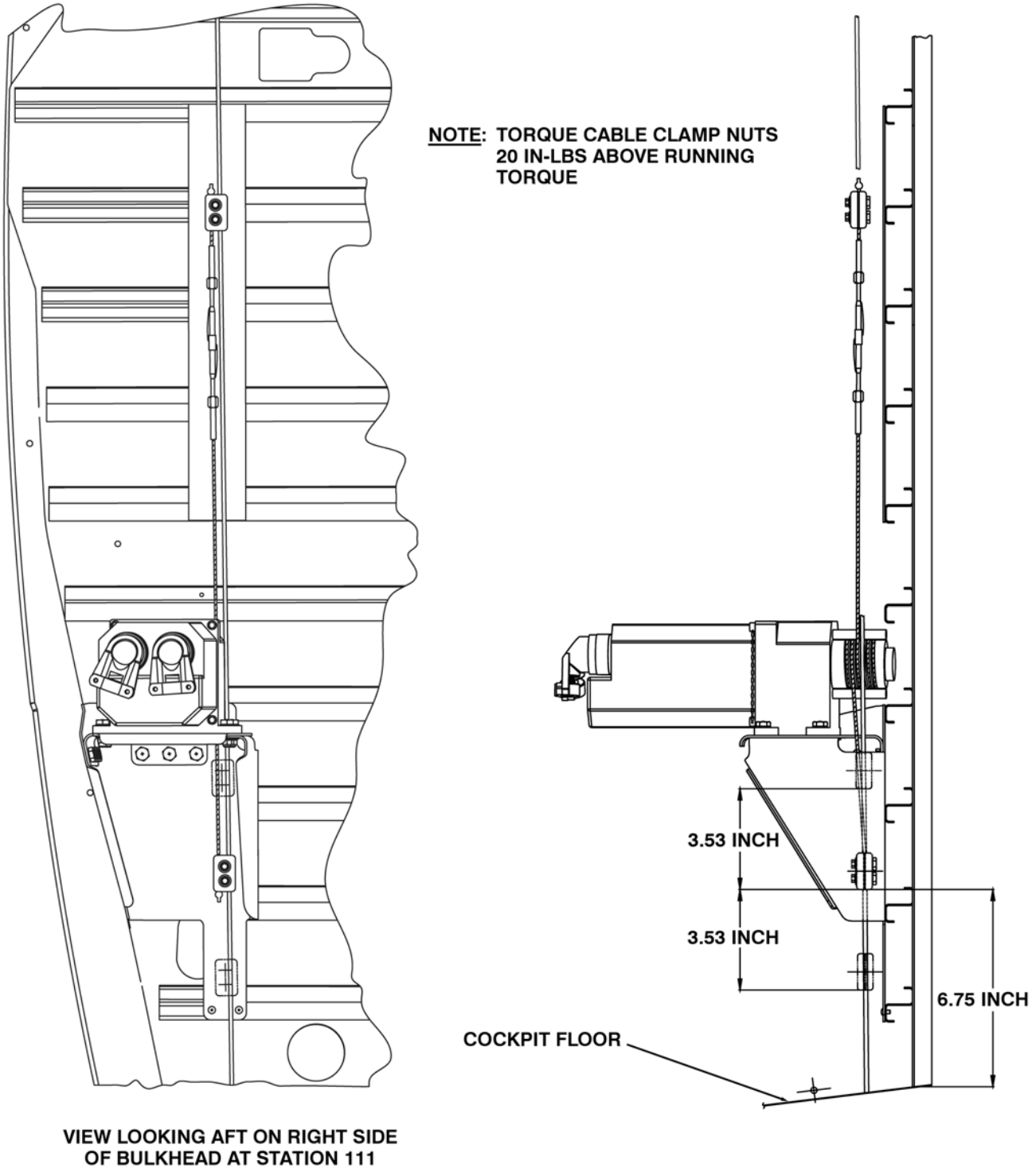
VIEW LOOKING DOWN & INBOARD ON LEFT SIDE
SHOWING INSTALLATION OF ELEVATOR SERVO
(RUDDER SERVO OMITTED FOR CLARITY)



SHOWING SECURING OF ELEVATOR SERVO BRIDLE CABLES TO ELEVATOR CONTROL CABLES
ELEVATOR IN FULL DOWN POSITION (CONTROL COLUMN FULL FORWARD)



Elevator Servo Capstan Installation
Figure 2



Aileron Servo Capstan Installation
Figure 3

k) Function test system. (Refer to 22-10-00, Automatic Flight Controls System (AFCS) (Mod 6/2303 and S.O.O. 6221) – General Data, Para 3.C. and Para 3.F.).

l) If rudder or elevator servo has been installed, install aft baggage extension floor panels and bulkhead panel at rear of rear baggage compartment. If aileron servo has been installed, install aileron cable covers in flight compartment and install Co-pilot seat.

B. Primary Servos Bridle Cable

Refer to Figure 4 for details.

1) Removal of Primary Servo Bridle Cable from Capstan

Note The Primary Servos (Rudder, Elevator and Aileron) can be replaced without removing the capstan and disturbing the cable rigging.

a) Ensure the aircraft electrical power is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Remove primary servo capstan bracket.

c) Disengage tabs of locking washer from cable keepers.

d) Detach cable keepers and locking washer from servo bracket. Discard locking washers.

e) Remove and retain bridle cable securing screws from capstan and detach bridle cables from capstan.

Note Inspect capstan cable grooves and bridle cables for wear.

2) Installation of Primary Servo Bridle Cable on Capstan

a) Ensure the aircraft electrical power is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Install bridle cables onto capstan and engage swaged cable terminal ends with capstan anchor points on capstan.

Note To ensure cable ball is captured, use screw MPC P/N 99945680061 only.

c) Install bridle cable securing screws, torque to 5 ± 0.5 in-lbs and lock wire with .02 diameter anneal monel wire. Ensure lockwire does not extend beyond the outer diameter of the capstan.

d) Wrap each bridle cable around capstan for approximately 2 wraps.

e) Assemble locking washers onto threaded end of cable keepers and install onto capstan. Torque cable keepers to 18±2 in-lbs. Engage one tab of locking washers with cable keeper flat.

Note Gap between cable keepers and capstan shall not exceed 0.04 inch.

f) Rig in accordance with Para 2.

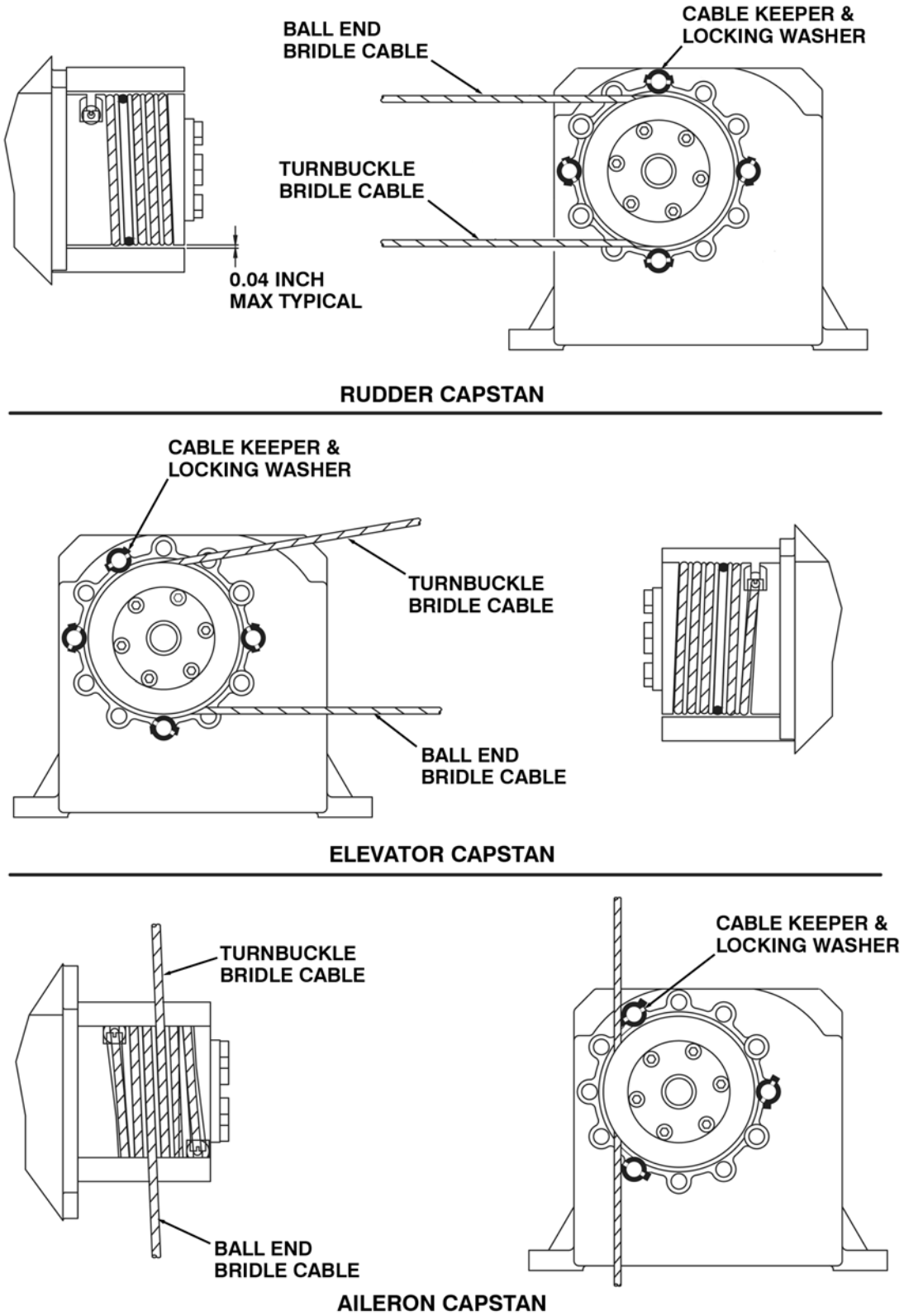
Note The aircraft must be allowed to stabilize at a temperature >55°F for a minimum of 12 hours prior to adjustment of the bridle cable tension.

Cable tension to be in accordance with Table 1 — Servo Bridle Cable Tension.

g) Check cable tensions. (Refer to 27-00-00, Flight Controls – General Data, Para 2.B.

h) Connect electrical connectors to servo.

i) Function test system. (Refer to 22-10-00, Automatic Flight Control System (Mod 6/2303 and S.O.O. 6221) – General Data, Para 3.C. and Para 3.F.).



Installation of Servo Bridle Cables
Figure 4

2. Adjustment/Test

Note Ensure aircraft conditioned at minimum temperature of 55°F (13°C).

A. Rig Rudder Servo Capstan

Refer to Figure 1 for details.

1) Ensure that the rudder control cable is correctly rigged. Refer to 27-20-00, Rudder Control System – General Data.

2) Hold rudder control in full right position.

3) Tension rudder control servo bridle cable tension to Table 1 — Servo Bridle Cable Tension. Install locking clips on turnbuckle.

4) Hold right rudder pedal in fully forward position and check that distance between the cable clamps is not less than 2 inches.

5) Operate rudder controls over full range and check that there is adequate clearance between cable clamps and structure.

B. Rig Elevator Servo Capstan

Refer to Figure 2 for details.

1) Ensure that the elevator control cable is correctly rigged. Refer to 27-30-00, Elevator Control System – General Data.

2) Hold elevator controls in full down position.

3) Tension elevator control servo bridle cable tension to Table 1 — Servo Bridle Cable Tension. Install locking clips on turnbuckle.

4) Hold at control column in full forward position and check that the distance between the cable clamps is not less than 1.5 inches.

5) Operate elevator controls over full range and check that there is adequate clearance between cable clamps and structure.

C. Rig Aileron Servo Capstan

Refer to Figure 3 for details.

1) Ensure that the aileron control cable is correctly rigged. Refer to 27-10-00, Aileron Control System – General Data.

2) Lock aileron controls in neutral position.

3) Tension aileron servo bridle cable tension to Table 1 — Servo Bridle Cable Tension. Install locking clips on turnbuckle.

4) Check that the distance between the lower edge of the servo cable clamp and floor is 6.75 inches.

5) Unlock aileron controls.

6) Operate aileron controls over full range and check that there is adequate clearance between cable clamps and structure.

Table 1 Servo Bridle Cable Tension

Item	Rudder	Elevator	Aileron
Bridle Cable Tension	25 lbs +5 lbs –0 lbs	25 lbs +5 lbs –0 lbs	15 lbs +2 lbs –0 lbs